Autonomous Campus CCTV Waste Surveillance Pipeline

8-Stage Computer Vision, Stratified Dynamic Auditing & BVG Dispatch Specification

Institution: Chhatrapati Shahu Ji Maharaj University (CSJMU), Kanpur • **Division:** Smart Campus Urban Planning & GIS Cell
Architecture Version: 2.4 (Production Institutional Build) • **Surveillance Scope:** 264-Acre Campus Core Precincts • **Target Engine:** YOLOv11-TACO + Centralized Inference

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1. Executive Summary & Operational Engineering Principles

Maintaining sanitation compliance across CSJMU's 264-acre campus requires continuous surveillance without imposing unrealistic network overhead or staffing costs. Standard commercial campus CCTV systems (CP PLUS, Hikvision, Dahua) cannot stream continuous 4K @ 30 FPS video feeds across campus core switches without severe bandwidth congestion. Furthermore, standard YOLO architectures natively downsample images to **640×640 tensors**, making 4K video decoding wasteful in both compute cycles and network capacity.

- **1. Ephemeral Sub-Stream / Snapshot Ingestion:** Instead of opening persistent 24/7 RTSP sessions, the system queries camera HTTP snapshot APIs (/ISAPI/Streaming/channels/101/picture) or pulls from RTSP sub-streams (Channel 102). Bandwidth consumption drops by over 99.8%.
- **2. Stratified Dynamic Random Auditing:** Static sweep times (e.g. 6 AM, 12 PM) lead to the 'anticipation effect,' where cleaning teams only tidy up minutes before scheduled sweeps. The pipeline implements a stratified pseudo-random scheduler that computes unpredictable audit timestamps inside 4 operational shift quarters (06:00 to 22:00) with a minimum 90-minute spacing buffer.
- **3. Centralized Datacenter Inference:** Because audits occur only 4 times daily across 10–30 cameras, central server inference processes all camera frames in **under 2.5 seconds total**, eliminating expensive and failure-prone outdoor edge hardware (Jetson) on camera poles.
- **4. End-to-End Closed Loop:** The pipeline connects optical detection directly to the CSJMU Ward Authority Matrix and BVG Sanitation Beat crews, completing work order generation, SLA tracking, and before-vs-after verification.

2. The 8-Stage CCTV Pipeline Architecture

The pipeline executes an autonomous 8-step sequence that transforms raw campus camera optics into verified sanitation ticket closures:

Stage	Pipeline Stage Name	Technical Process & Mechanism	Output / Artifact
Stage 1	**CCTV Data Collection**	Stratified random trigger activates sweep across campus camera registry across 4 shift windows.	Triggered Camera Manifest
Stage 2	**Data Ingestion & Storage**	HTTP GET ISAPI/ONVIF snapshot capture (~180 KB), letterbox resize, face/license-plate privacy blur.	Clean 640x640 Input Buffer
Stage 3	**AI Computer Vision**	YOLOv11-TACO inference (6 classes), confidence gate (≥0.70), volumetric & weight estimation.	Bounding Box, Class, Weight (kg)
Stage 4	**Geotagging & GIS**	Camera ID lookup against campus spatial database; tags 2D floor coordinates and WGS84 GPS.	Geolocated Telemetry Package
Stage 5	**LLM Structured Parsing**	Multimodal LLM synthesizes telemetry into municipal title, natural language summary, and SOP action plan.	Structured Work Order JSON
Stage 6	**Authority Routing Engine**	Spatial query resolves: Zone → Ward → Department → Supervisor → BVG Sanitation Team.	Assigned Team & SLA Deadline

Stage 7	**Field Dispatch**	Automated incident creation in database, push notification to field beat leader's handheld terminal.	Active Ticket (DISPATCHED)
Stage 8	**Cleaning Verification**	Post-cleaning camera re-capture or worker handheld photo comparison. AI confirms clean state.	Ticket Resolved & Audit Logged

3. Campus Surveillance Registry: Institutional Hardware Specs

The table below details the active CCTV units configured in the CSJMU production registry, reflecting standard institutional 2 MP and 1.3 MP deployments:

Camera ID	Location & Campus Zone	Resolution & FPS	Stream Protocol	Channel	Target YOLO Tensor
CAM-UIET-01	UIET Block-3 Canteen Concourse	1080p (1920×1080) @ 15 fps	SUB_STREAM_RTSP	Ch 102	640×640 px
CAM-LIB-02	Central Library Plaza Entrance	1080p (1920×1080) @ 15 fps	SUB_STREAM_RTSP	Ch 102	640×640 px
CAM-WS-03	Central Workshop North Drain	720p (1280×720) @ 15 fps	SUB_STREAM_RTSP	Ch 102	640×640 px
CAM-HST-04	Ramanujan Hostel Mess Quad	1080p (1920×1080) @ 15 fps	SUB_STREAM_RTSP	Ch 102	640×640 px
CAM-STAD-05	Stadium Pavilion South Bleachers	720p (1280×720) @ 15 fps	SUB_STREAM_RTSP	Ch 102	640×640 px
CAM-ADMIN-06	VC Secretariat VIP Porch	1080p (1920×1080) @ 15 fps	HTTP_SNAPSHOT_ISAPI	Ch 101	640×640 px

4. Network Bandwidth & Ingestion Economics

The design decision to utilize HTTP snapshot queries and sub-streams rather than continuous main-stream video is grounded in the following mathematical reality:

Ingestion Strategy	Bitrate / Size Per Cam	Total Daily Data (20 Cams)	Switch Impact & Risk
Continuous 4K Streams (3840×2160 @ 30fps H.264)	~12.0 Mbps per stream	**~2,592 GB / day** (240 Mbps continuous)	**Severe Congestion:** Saturates campus backbone, drops VoIP/academic packets, requires heavy GPU decoding servers.
Continuous 1080p Sub-Streams (1920×1080 @ 15fps H.264)	~1.8 Mbps per stream	**~388 GB / day** (36 Mbps continuous)	**Moderate Overhead:** Substantially lower, but still wastes bandwidth streaming during unoccupied hours.
CSJMU Ephemeral HTTP Snapshots (Single 1080p JPEG @ 4 sweeps/day)	~180 KB per capture query	**~14.4 MB / day** (0.00017 Mbps avg)	**Optimal Institutional Fit: 99.99% bandwidth savings.** Zero network contention. Sub-second queries directly into YOLO tensor memory.

5. Stratified Dynamic Random Auditing Scheduler

To prevent facilities crews from anticipating audits, CSJMU divides the active campus day (06:00 to 22:00) into four non-overlapping 4-hour operational quarters. A pseudo-random generator computes trigger timestamps with non-clustering boundary buffers:

Quarter	Shift Window	Audit Focus & Critical Targets	Trigger Offset Algorithm
1st Sweep	06:00 – 10:00	Overnight waste accumulation, hostel mess dumping, pre-class concourses.	$t \in [06:25, 09:35]$ (Gaussian / uniform offset)
2nd Sweep	10:00 – 14:00	UIET canteen peak, lunch kiosk overflow, open lawn takeaway wrappers.	$t \in [10:25, 13:35]$ (≥90 min from Sweep 1)
3rd Sweep	14:00 – 18:00	Post-afternoon dispersal, sports bleachers, library walkway bins.	$t \in [14:25, 17:35]$ (≥90 min from Sweep 2)
4th Sweep	18:00 – 22:00	Hostel dinner mess exits, evening study walkways, gate bins before night lock.	$t \in [18:25, 21:35]$ (≥90 min from Sweep 3)

6. AI Detection Classes & Volumetric Density Model

The model is fine-tuned on municipal waste datasets (TACO / ZeroWaste) to recognize 6 operational states. Each class is mapped to bulk density coefficients to convert visual volume into estimated weight in kilograms:

Detection Class	Priority	Bulk Density	SLA Target	Standard Operating Dispatch Action
blocked_drain	**CRITICAL**	160 kg/m³	**30 min**	Deploy desilting squad with suction pumps and rods to avert campus storm flooding.
overflow	**HIGH**	120 kg/m³	**45 min**	Dispatch 2-person BVG crew with 240L wheelbarrow. Clear overflow and disinfect perimeter.
litter_pile	**HIGH**	90 kg/m³	**60 min**	Dispatch manual sweeping squad with litter pickers and heavy-duty polythene bags.
bin_full	**MEDIUM**	110 kg/m³	**90 min**	Schedule compactor truck beat round to empty bin before overflow threshold.
bin_partial	**LOW**	100 kg/m³	Next Shift	Logged in daily analytics. No emergency dispatch required.

bin_empty	**NORMAL**	N/A	Audit Pass	Recorded as positive cleanliness compliance for zone supervisor scorecard.

7. Municipal Authority Hierarchy & BVG Workforce Dispatch

A critical innovation in CSJMU's system is linking AI detection directly to a real-world municipal responsibility chain. Detections do not end as passive dashboard markers—they trigger legally enforceable work orders:

Campus Precinct	Ward Designation	Departmental Owner	Beat Supervisor	Assigned Field Crew
UIET Academic Block	Ward 4 – Academic Division	Estate Office / Academic Maintenance	Er. Mahendra Gupta *(Sanitation Inspector)*	BVG Beat Squad Alpha (2 Sweep + 1 Cart)
Central Library & Admin	Ward 2 – Central Secretariat	Registrar's Office Facilities Cell	Mr. R. K. Srivastava *(Assistant Registrar)*	BVG Admin Flying Squad (VIP Beat)
Mechanical Workshops	Ward 5 – Industrial & Utilities	Works & Infrastructure Dept	Er. Suresh Verma *(Works Supervisor)*	Utility & Drain Clear Crew (Heavy Rodding)
Ramanujan Hostels	Ward 7 – Student Residences	Chief Warden Office	Dr. Pradeep Mishra *(Hostel Sanitation Supv)*	BVG Hostel Sanitation Beat (Mess Clearing)

8. Micro-Litter Detection Strategy: SAHI vs. Receptacle ROI

When operating on 1080p and 720p streams, individual plastic cups or wrappers cover fewer than 20×20 pixels in wide outdoor shots. Standard YOLO downsampling can cause these objects to be missed. The pipeline employs a two-tier strategy:

- **Strategy A: Macroscopic Priority (Default):** The model prioritizes overflow, litter piles, and clogged storm drains, which represent >92% of municipal health risks and are reliably detected at 640×640.
- **Strategy B: Static Receptacle ROI Cropping:** For fixed cameras focused on dustbins (e.g. CAM-UIET-01), the engine crops a predefined bounding polygon (e.g. 5-meter radius around the bin) and passes this crop directly to YOLO at full resolution without global image downsampling.
- **Strategy C: SAHI (Slicing Aided Hyper Inference):** For wide plaza shots (CAM-LIB-02), the 1080p snapshot is dynamically sliced into overlapping 640×640 tiles with 20% overlap, followed by Non-Maximum Suppression (NMS) merging. This yields 4K-level detection accuracy while consuming only 180 KB of network transfer.

9. Decision Engine: De-Duplication Gate & Verification Loop

To avoid dispatch spamming (e.g. creating duplicate tickets on successive sweeps for the same pile before the cleaning team arrives), the Decision Engine runs a spatial deduplication filter:

Decision Gate Logic:

```
passes_confidence = (confidence ≥ 0.70)
is_significant = (detection_class not in ['bin_empty', 'bin_partial'])
is_duplicate = is_duplicate_incident(camera.zoneId, detection_class, camera.id)
should_dispatch = passes_confidence AND is_significant AND NOT is_duplicate
```

Stage 8 Verification: When a cleaning worker marks a work order complete via the CSJMU field mobile app, they upload an 'After' photograph. The engine verifies the resolution by checking for the absence of the previously flagged class, recording before-and-after photographic evidence in the permanent campus sanitation ledger.

Technical Specifications Approved By:
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